Removable the limit exists; the fault is a single value, so redefining $f(a)$ makes the function continuous

- **Hole** $f(a)$ is undefined — $(x^2 - 1)/(x - 1)$ at $x = 1$: limit 2, no value
- **Misplaced value** $f(a)$ exists but differs from the limit — the point sits off the curve

Non-removable the limit does not exist; no single value of $f(a)$ can fix it

- **Jump** both one-sided limits exist but differ — $\lfloor x \rfloor$ at $x = 2$: 1 from the left, 2 from the right
- **Infinite** at least one one-sided limit is $\pm\infty$ — $1/x$ at $x = 0$
- **Oscillating** the limit fails to exist because the function oscillates without settling — $\sin(1/x)$ at $x = 0$

the limit exists: one value repairs it the limit does not exist: nothing repairs it